

Signal2Hand: Sensor Modality Translation from Body-Worn Sensor Signals to Hand-Depth Images

Yuki Kubo
NTT Corporation
Japan
yukikubo@acm.org

Buntarou Shizuki
University of Tsukuba
Japan
shizuki@cs.tsukuba.ac.jp

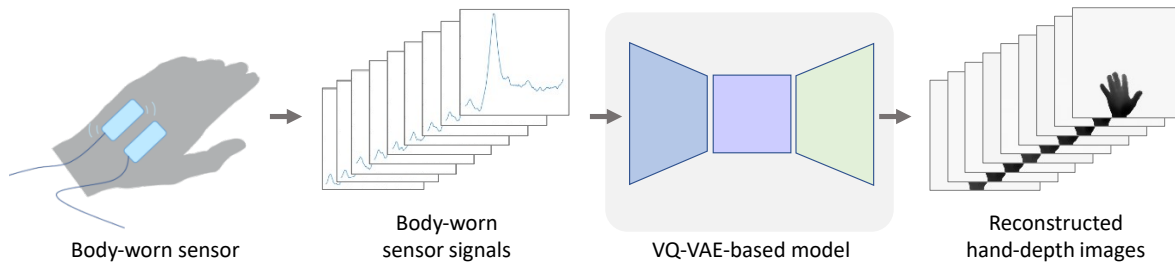


Figure 1: Overview of Signal2Hand. Signal2Hand is hand reconstruction method that directly reconstructs hand-depth images from body-worn sensor signals.

Abstract

We present Signal2Hand, a method that reconstructs hand-depth images from body-worn sensor signals. To do this, we trained VQ-VAE using a pair of body-worn sensor signals and a hand-depth image. We conducted a preliminary study of the method and evaluated the reconstruction results using four evaluation metrics.

CCS Concepts

• **Human-centered computing** → *Interaction techniques*.

Keywords

Hand-depth image reconstruction, active bio-acoustic sensing

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1 Introduction

Hand reconstruction has various applications in the fields of Human-Computer Interaction (HCI) and Augmented Reality (AR) / Virtual Reality (VR); one such application is the visualization of expert skills [Han et al. 2020; Karunratanakul et al. 2023]. However, current hand reconstruction methods use cameras, making them not suitable in situations where occlusion occurs or lighting conditions vary. The motivation for our work is to realize hand reconstruction

that is not affected by these effects. In addition, we want to achieve this in a form that minimizes the hindering of finger movements, unlike traditional data gloves.

In this work, we present Signal2Hand, a method that directly reconstructs hand-depth images from body-worn sensor signals. The method first trains a Vector Quantized Variational Autoencoder (VQ-VAE) [Van Den Oord et al. 2017] using a pair of body-worn sensor signals and a hand-depth image. After the training, the method can reconstruct directly the corresponding hand-depth image by inputting the signals to the trained model. In the following sections, we describe the method including the data and model used in it, and the quantitative results of reconstruction in a preliminary study.

2 Method

Signal2Hand (Figure 1) directly reconstructs hand-depth images from body-worn sensor signals and thus is unaffected by occlusion or varying lighting conditions. For the sensor signal, we use bio-acoustic signals of the hand [Kubo et al. 2019]. These signals vary in accordance with hand postures and can be acquired by active bio-acoustic sensing. In the sensing method, the sensor emits ultrasound from one piezo element (serving as a speaker) attached to the back of the hand, which travels through the hand and along its surface. We then obtain an ultrasound (i.e., sensor signals) with another piezo element (serving as a microphone) attached to the back of the hand. Different from the previous work, we use linear frequency cepstral coefficients (LFCCs) [Amesaka et al. 2022] to generate features. Specifically, we first calculate LFCCs in the range of 20–40 kHz as a 200-element vector for the frequency response obtained by fast Fourier transform (FFT) for each frame at approximately 20 Hz; we use filter banks with a bandwidth of 200 Hz and a center frequency ranging from 20 to 39.9 kHz with an interval of 100 Hz. To reduce noise, we generate the 200-element feature vector by averaging two-frame LFCCs. Then, we cut off high-frequency components;

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as a result, the number of elements in the vector becomes 192. In addition, we normalize the vector using a min-max normalization. Finally, we augment the one-dimensional vector to a 16-dimensional feature with a complete copy. As a result, the dimension of a feature of bio-acoustic signals is $192 \times 16 \times 1$.

To collect hand-depth images, we used an Azure Kinect DK (Microsoft Corp.) with Kinect Azure SDK 1.4.1. The mode of the camera was NFOV Unbinned. First, we remove near and distant pixels from the acquired depth images (640×576 pixels) to isolate the hand components. Next, we extract the hand region at a size of 256×256 pixels and then resize it into a 128×128 pixel image to match the output size of the network model. We then normalize the image so that each pixel has an intensity value between 0 (black) and 255 (white).

The structure of the VQ-VAE is designed so that the input to the encoder is the sensor signal (i.e., a 16-dimensional feature), and the decoder output is a hand-depth image. The $192 \times 16 \times 1$ feature to the encoder becomes $16 \times 16 \times 256$ data with four Conv2D layers. The decoder is structured with four-layer groups, each having three transposed Conv2D layers. It upscales a $16 \times 16 \times 1$ latent variable obtained from the codebook of the VQ-VAE by four times and reconstructs a $128 \times 128 \times 1$ hand-depth image as output. We define a latent embedding space $e \in R^{K \times D}$ with $K = D = 256$. Our VQ-VAE-based model uses the mean squared error (MSE) between the reconstructed hand-depth image and its corresponding ground truth as the reconstruction loss. The other losses are the same as those of the VQ-VAE. During training, we use a batch size of 4 and update the weights using the Adam optimizer with a learning rate of 0.001.

3 Preliminary Study

We evaluated the quality of the reconstructed hand-depth images with five image quality evaluation metrics. We recruited 10 participants (7 males and 3 females) ranging in age from 25 to 44 (M: 30.5, SD: 5.85). We first asked participants to attach the two piezo elements to the back of their right hand. The locations were 10 mm from the metacarpal (MCP) joints (the knuckle between the hand and finger). The distance between the piezo elements was 10 mm, which was chosen according to the results reported in [Kubo et al. 2019]. During this study, the participants placed their right forearm

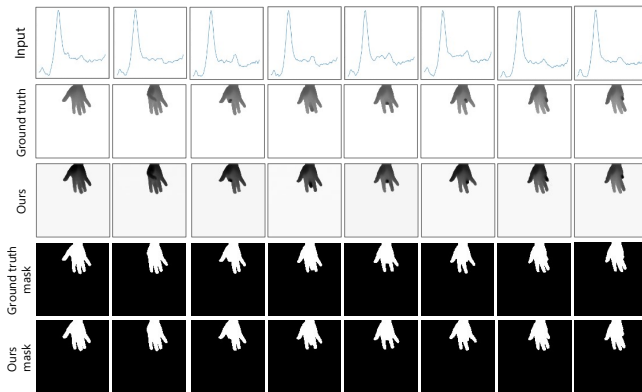


Figure 2: Qualitative results of preliminary study.

Table 1: Quantitative results of preliminary study.

	IoU $\uparrow$	L1 $\downarrow$	LPIPS (AlexNet/VGGNet) $\downarrow$	MS-SSIM $\uparrow$
Ours	0.780	0.062	0.054 / 0.062	0.916

on a desk to prevent fatigue. In a session, we asked the participants to repeatedly bend individual fingers in the instructed order (thumb, index finger, ..., pinky finger, thumb, ...) and at different speeds (neutral and slow). We collected data at neutral and slow speeds, 200 and 300 frames in a session. In summary, we collected the data of 10 participants $\times$ 4 sessions $\times$ 500 frames = 20,000 frames.

We conducted a leave-one-session-out cross-validation for each participant for the evaluation. In each training, we used 1,500 pieces of data for training and reserved the remaining 500 for testing. Four image quality metrics [Karunratanakul et al. 2023] were used to assess the reconstruction performance.

The results are shown in Table 1. For the four metrics, except for pixel-wise L1, the values were in the range of those reported (IoU: 0.779-0.929, L1: 0.018-0.051, LPIPS: 0.071-0.173, MS-SSIM: 0.831-0.902) in the previous work [Karunratanakul et al. 2023]. Figure 2 shows the reconstructed hand-depth images for the qualitative evaluation. The method appears to output images as clear as the depth images used as the ground truth. Therefore, it is suggested that the model of the method can reconstruct hand-depth images with a high reconstruction quality.

4 Conclusion

We presented Signal2Hand, a method that reconstructs hand-depth images from body-worn sensor signals. Signal2Hand reconstructs the images from the signals with only a trained VQ-VAE using pairs of bio-acoustic signals acquired from sensors worn on the back of the hand and hand-depth images. Preliminary study results showed that a hand-depth image could be reconstructed directly from the body-worn sensor signals.

In the future, we will compare and evaluate the results of hand reconstruction using other methods and our method. We will also evaluate the reproducibility of our method when the sensors are re-worn.

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